# Full Stack Development with MERN

# Project Documentation

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## 1. Introduction

- **Project Title: Streamlining Ticket Assignment For Efficient Support Operations**
- **Team Members:**
  - **Dhanasekar P – Full Stack Developer / Project Lead**
  - **Team Member 2 – Backend Developer**
  - **Team Member 3 – Frontend Developer**

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## 2. Project Overview

**Purpose**

**The project aims to develop an automated ticket assignment system that improves support operations by reducing manual effort, minimizing errors, and ensuring faster issue resolution.**

**Features**

- **User Registration & Login**
- **Ticket Creation & Tracking**
- **Automated Ticket Assignment**
- **Role-Based Access Control (RBAC)**
- **Dashboard for Monitoring**
- **Notifications & Status Updates**

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## 3. Architecture

**Frontend (React)**

- Built using React.js with component-based architecture
- Pages include Login, Register, Dashboard, Ticket Form
- Uses Axios for API communication
- State managed using React Hooks

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**Backend (Node.js & Express.js)**

- RESTful APIs built using Express.js
- Handles:
  - Authentication
  - Ticket management
  - Assignment logic
- Middleware used for validation and security

---

**Database (MongoDB)**

## Collections:

## User Collection

```
{
 "name": "string",
 "email": "string",
 "password": "string",
 "role": "string"
```

```
}
```

**Ticket Collection**

```
{

 "title": "string",

 "description": "string",

 "category": "string",

 "status": "Open/In Progress/Closed",

 "assignedTo": "userId"

}
```

---

## 4. Setup Instructions

**Prerequisites**

- Node.js
- MongoDB
- Git

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**Installation Steps**

```
# Clone repository

git clone
   https://github.com/your-repo/project.git
```

```
# Navigate to project
cd project

# Install frontend dependencies
cd client
npm install

# Install backend dependencies
cd ../server
npm install
```

---

## 5. Folder Structure

**Client (React Frontend)**

```
client/
├── src/
│   ├── components/
│   ├── pages/
│   ├── services/
│   └── App.js
```

└── package.json

---

**Server (Node Backend)**

server/

├── models/

├── routes/

├── controllers/

├── middleware/

└── [server.js](server.js)

---

## 6. Running the Application

**Frontend**

```
cd client
npm start
```

**Backend**

```
cd server
npm start
```

---

## 7. API Documentation

**Register User**

- POST `/api/register`

**Request Body:**

```
{
 "email": "user@example.com",
 "password": "123456"
}
```

---

**Login User**

- **POST** `/api/login`

---

**Create Ticket**

- **POST** `/api/tickets`

---

**Get Tickets**

- **GET** `/api/tickets`

---

**Update Ticket**

- **PUT** `/api/tickets/:id`

---

## 8. Authentication

- Uses JWT (JSON Web Token) authentication
- Token generated during login
- Stored in browser local storage
- Middleware verifies token for protected routes

- **Role-based authorization ensures proper access**

---

**9. User Interface**

**Includes:**

- **Login Page**
- **Registration Page**
- **Dashboard**
- **Ticket Creation Form**
- **Ticket Status Tracking**

***(Attach screenshots or GIFs here in your report)***

---

**10. Testing**

- **Manual testing for UI and functionality**
- **API testing using Postman**
- **Tested modules:**
  - **Authentication**
  - **Ticket creation & assignment**
  - **Role-based access**

---

**11. Screenshots / Demo**

- **Add screenshots of:**
  - **Login Page**
  - **Dashboard**
  - **Ticket Creation**
  - **Ticket Assignment**
- **Demo Link: (Add your project demo link)**

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## 12. Known Issues

- Limited automation rules
- No AI-based classification
- UI can be further enhanced

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## 13. Future Enhancements

- AI-based ticket classification
- Chatbot integration
- Advanced analytics dashboard
- Mobile application support

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## Conclusion

This MERN stack project successfully implements a full-stack solution for automating ticket assignment, improving efficiency, and enhancing the overall support system experience.

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